Do Pose Estimators Need Extra Machinery to Know When They Fail? A Held-Out Comparison of One-Pass Signals, MC-Dropout and Learned Heads

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Abstract

Safety-relevant uses of 2D human pose estimation need to know which keypoints are wrong. Common answers add cost: repeated stochastic passes through MC-Dropout, or a trained uncertainty head. We ask how much these add over signals that a top-down model already produces. On three published models (SimCC with ViPNAS-MobileNetV3 and ResNet-50, and a ResNet-50 heatmap model), with every fitted parameter restricted to a calibration split of COCO val2017 and every reported value taken from a separate test split, the spread of the predicted coordinate distribution detected keypoint failures with AUROC 0.913 and 0.914 on SimCC, against 0.820 and 0.873 for 20-pass head MC-Dropout. On the heatmap model, peak confidence reached 0.906 against 0.827. MC-Dropout stayed below these one-pass signals at all five dropout rates tested. Using the spread to scale a conformal region shortened the mean radius on SimCC and raised coverage of occluded joints, but enlarged the radius on the heatmap model. Rescoring instances with the same signal changed AP by about +0.26 points at most, a gain close to its variation across folds. Fine-tuning a learned uncertainty head with heteroscedastic, calibration and OKS terms did not beat the spread of the same model: on ViPNAS-MobileNetV3 over three seeds, learned σ minus spread was -0.008 in failure AUROC, and the AP difference from a control fine-tuned with the same schedule was -1.07 points. On ResNet-50, one seed, σ also trailed spread and the AP change was within fold variation. Trained from ImageNet weights on a shortened schedule, SOCA-PRNet-34 differed from a plain ResNet-34 by +0.44 AP points, less than the variation across folds, in one run per backbone.

Keywords: human pose estimation, uncertainty estimation, failure detection, MC-Dropout, coordinate classification, conformal prediction

1. Introduction

A pose estimator that returns a wrong joint with no warning is a problem wherever the output drives a decision, as in rehabilitation, ergonomics or human–robot interaction. Occluded joints are the clearest case. On our test split, the SimCC ViPNAS-MobileNetV3 model places 0.931 of visible joints within the PCK@0.1 tolerance but only 0.663 of occluded ones.

The usual remedies add something to the model. MC-Dropout [1] repeats inference with dropout active, and heteroscedastic heads [2] add parameters and a training run. Recent pose work has moved towards richer probabilistic outputs, including residual likelihoods [3], calibration against object keypoint similarity [4], and probabilistic formulations of the output map [5]. Yet a coordinate-classification head [6] already outputs a full 1D distribution for each axis, and a heatmap head a 2D one. Whether the additions improve on these existing outputs is an empirical question, and its answer decides whether the added cost is justified, especially on lightweight models.

We test this directly, with four questions fixed in an analysis plan before any result was inspected (Section 3.1):

RQ1. On published pretrained models, do one-pass signals match 20-pass MC-Dropout at detecting keypoint failure and occlusion?

RQ2. Does fine-tuning a learned uncertainty head improve on those signals, and what does each component add?

RQ3. What does the added machinery cost in AP, parameters, FLOPs and latency?

RQ4. Does uncertainty-based rescoring raise COCO AP on held-out images?

Our contributions are:

- a held-out comparison of eight uncertainty signals across three models, with paired image-level bootstrap intervals;
- a sensitivity analysis showing that the gap to MC-Dropout does not depend on the planned dropout rate or on flip testing;
- an ablation of a learned head on the same backbones and schedule, with the planned arms reported whichever way they turned out;

- measured latency, showing that T-pass head MC-Dropout costs far less than T forward passes, so the argument against it cannot rest on latency alone; and
- a statement of the implementation defects we found and fixed, several of which silently changed results while the code still ran (Appendix A).

2. Related work

Pose estimation heads. SimCC [6] treats each coordinate as a classification over sub-pixel bins, and RTMPose [7] builds a real-time pipeline on the same idea. Heatmap models such as SimpleBaseline [8] regress a 2D map. PCT [9] represents a pose as compositional tokens, the inspiration for the token mixer we evaluate. ViPNAS [10] supplies our lightweight backbone, and SOCA-PRNet [11] adds attention to a PoseResNet, drawing on global context [12] and coordinate attention [13].

Uncertainty in pose estimation. RLE [3] learns a residual likelihood for regression. Gu et al. [4] study the mismatch between predicted confidence and OKS, and train a confidence target aligned with OKS, which we reuse as one component of the learned head. ProbPose [5] develops a probabilistic formulation of the keypoint output, and RTMO [14] is a one-stage model with coordinate classification. More recent work covers post-hoc predictive distributions for YOLO-Pose [15], evidential pose estimation [16], and uncertainty-aware models for multi-person [17] and motor-ability-specific [18] settings. Our study differs in its question: it does not propose a new estimator, but measures what such machinery adds over existing outputs.

Evaluating uncertainty. Sparsification and AUSE come from optical flow [19]. Comparisons of regression uncertainty metrics [20, 21] motivate reporting several metrics, not one. We use Platt scaling [22] and report ECE [23] after calibration. Split conformal prediction [24] has been applied to keypoints for object pose [25]. Benchmark variance [26] motivates our paired intervals and repeated seeds.

3. Methods

3.1. Analysis plan and data

The research questions, metrics and arms were written down before the first benchmark run, and later changes were logged with their reasons (`paper/ANALYSIS_PLAN.md`). The dropout-rate sweep, the no-flip dumps,

the widened rescoring grids and one training arm (Section 3.5) were added after seeing results, and we label them exploratory.

Training uses COCO [27] train2017 only. COCO val2017 is split once at image level, with a fixed seed, into a calibration part of 1,500 images and a test part of 3,500 images. Platt parameters, fusion weights, rescoring parameters and conformal quantiles are fitted on the calibration part. Every reported metric comes from the test part, except the reproduction check in Table 1, which uses all images to match the published protocol. Reported checkpoints are those of the final epoch, so no checkpoint is chosen using evaluation images.

Keypoint-level metrics use ground-truth boxes, since occlusion flags need a correspondence with annotations. The test part contains 47,149 labelled keypoints from 1,644 images with labelled people: 41,442 visible (v=2) and 5,707 occluded (v=1). Unlabelled keypoints are excluded.

3.2. Models

We use the published MMPose [28] checkpoints of SimCC with ViPNAS-MobileNetV3 (input 256×192), SimCC with ResNet-50, and a SimpleBaseline ResNet-50 heatmap model. Flip testing is on unless stated otherwise. Detector-box AP uses torchvision's Faster R-CNN R50-FPN v2 person boxes. Their person box AP is 59.5 (AP_{50} 86.2), whereas the file behind the published MMPose numbers is named for AP 56.4, so the comparison with published AP in Table 1 is approximate.

3.3. Uncertainty signals

For a SimCC joint, let p_x and p_y be the softmax distributions over bins, with bin centres b_i in input pixels.

- **Peak confidence**: the model's default score, derived from the raw logits. For SimCC it is unbounded above by construction, so it is not a probability.
- **Peak probability**: $\min(\max_i p_x(i), \max_i p_y(i))$.
- **Distribution spread**: $u_{\text{dist}} = \sqrt{\text{Var}_{p_x}[b] + \text{Var}_{p_y}[b]}$. On the heatmap model, the analogous spatial standard deviation around a temperature-sharpened soft-argmax.
- **Entropy**: the mean of the two axes' Shannon entropies, each normalised to $[0, 1]$.

- **Flip disagreement**: the distance between the decoded coordinates of the original and flipped views.
- **MC-Dropout**, T=20: the coordinate standard deviation over stochastic passes, with *argmax* and *soft-expectation* decoders, both kept so that the baseline is not handicapped by the decoder choice.
- **Fusion**: logistic regression on peak confidence, spread, entropy and flip disagreement, fitted on the calibration part only.
- **Learned σ and learned OKS**: outputs of the trained head (Section 3.4), reported only for arms that trained them.

MC-Dropout is applied in the *head* only, on the features of one unflipped pass: at the feature layer before the SimCC classifiers (and in the mixer, when present), and as spatial dropout after each deconvolution on the heatmap model. The backbones contain no dropout. The pretrained models were not trained with this dropout, so on them it is a perturbation rather than a trained posterior approximation. The `base` arm in Section 3.5 addresses this by training with the dropout active.

3.4. Learned head

The head adds a per-joint log-variance s_k and an OKS logit, with an optional token mixer. With D the per-axis KL divergence summed over bins, the loss adds to SimCC's KL term

$$\mathcal{L}_{\text{het}} = \tfrac{1}{2} e^{-s}\, D + \tfrac{1}{2} s \qquad \text{attenuation [2],} \tag{1}$$

$$\mathcal{L}_{\text{cal}} = |r|\, e^{-s/2} + \tfrac{1}{2} s \qquad \text{Laplace NLL on the decoded residual } r, \tag{2}$$

$$\mathcal{L}_{\text{oks}} = \text{BCE}\big(z,\ e^{-d^2/(2A(2\kappa)^2)}\big) \qquad \text{OKS target, after [4],} \tag{3}$$

where r and s are in units of the target width σ_0, d is the decoded error in input pixels, κ the COCO falloff constant of the joint, and A the median train2017 person area in crop pixels. The optional adaptive target is an equal mix of the original label and a Gaussian whose width is scaled by $e^{s/2}$, clamped to $[0.7, 2.5]$, with s detached. Weights are 0.5, 0.2 and 0.2. The token mixer starts as an identity through a zero-initialised residual, so it cannot degrade the pretrained model at the start of training.

3.5. Training arms

Each arm fine-tunes the published checkpoint for 20 epochs on one GPU: AdamW, batch size 64, learning rate 3×10^{-4} for new modules with multipliers 0.1 (backbone) and 0.2 (deconvolution head), cosine decay, mixed precision, and gradient clipping at norm 1. The arms accumulate components: `base` (KL only, the control), `het` ($+\mathcal{L}_{\text{het}}, \mathcal{L}_{\text{cal}}$), `het_oks` ($+\mathcal{L}_{\text{oks}}$), `mix` (+ token mixer) and `full` (+ adaptive target).

One arm is exploratory. At an interim check on all val2017 images, `full` had AP 0.698 at epoch 5 against 0.711 for `base`, with gradient norms of about 100 before clipping. We suspected that e^{-s} in $\mathcal{L}_{\text{het}}$, which grows as accurate joints reach a low s, multiplies the pose gradient. We kept all planned arms and added `full_sg`, identical to `full` except that D is detached in $\mathcal{L}_{\text{het}}$. Because an evaluation-set number motivated it, its results are reported separately and do not replace `full`.

The SOCA-PRNet comparison trains two backbones from ImageNet weights with the same SimCC head and the KL loss only: a plain ResNet-34 and SOCA-PRNet-34, which is the same ResNet-34 with SOCA attention and, as part of its design, dropout of 0.05 inside the backbone. Both use AdamW with learning rate 5×10^{-4} and weight decay 10^{-4}, batch size 64, a 500-iteration linear warm-up, and 60 epochs with the learning rate divided by 10 after epochs 48 and 57. This is shorter than the 210-epoch schedule of the published models, so neither result can be compared with published numbers. Each backbone was trained once.

3.6. Metrics

A keypoint *fails* when its error, normalised by the square root of its box area, is at least 0.1. Primary metrics are failure AUROC, AUSE on normalised error (retained fractions 0.1 to 1), and occlusion AUROC (v=1 against v=2). AUROC handles ties by average rank, and sparsification breaks ties randomly. Secondary metrics are Spearman ρ with error, ECE after Platt scaling, and split-conformal regions at 90% target coverage, where a radius is $\hat{q}\,u$ with $\hat{q}$ the calibration quantile of error divided by u.

Intervals are percentile 95% over 1,000 image-level bootstrap resamples, with the same resamples for all signals, so paired differences are valid. We call a difference supported only if its interval excludes zero. AP comes from pycocotools with the MMPose score mode and OKS-NMS. Rescoring differences are the mean paired difference over 10 disjoint folds of the test part.

Table 1: AP on all val2017 images, with default instance scores. †MMPose model zoo values, not re-measured, obtained with a different detector.

Model	AP (GT boxes)	AP (det. boxes)	AP_{50} (det.)	AR (det.)	Published AP†
SimCC-ViPNAS-MBV3	0.711	0.691	0.882	0.748	0.695
SimCC-ResNet-50	0.739	0.720	0.899	0.774	0.721
Heatmap-ResNet-50	0.737	0.719	0.900	0.768	0.718

3.7. Instance rescoring

The default instance score is the box score times the mean confident keypoint score. We compare exponential decay of keypoint scores $c_k e^{-\alpha u_k}$, with u_k normalised by its mean, and an instance score of the box score times the mean closed-form expected OKS under isotropic Gaussian error, $(1 + (\gamma\sigma)^2/(2c))^{-1}$ with $c = A(2\kappa)^2$. Parameters α and γ are selected on the calibration part. Expected OKS is omitted for the heatmap model because its spread is not in pixel units.

3.8. Cost measurement

Latency uses batch size 1, 50 warm-up runs and 300 timed runs, reporting the median, on an RTX 5060 (8 GB) and on one CPU thread of a Core Ultra i7. No training job was running on the GPU, though desktop applications were. FLOPs and parameters come from the unmodified MMPose models where marked "plain".

4. Results

4.1. Reproduction check

Table 1 shows that our pipeline reproduces published AP to within 0.4 points on detector boxes (0.720 vs 0.721, 0.691 vs 0.695, and 0.719 vs 0.718), despite the different detector.

4.2. RQ1: one-pass signals against MC-Dropout

Table 2 is the main zero-shot result. On SimCC, distribution spread had the highest failure AUROC and lowest AUSE on both backbones. Its paired AUROC gain over soft MC-Dropout was +0.093 [0.086, 0.100] on ViPNAS-MobileNetV3 and +0.041 [0.036, 0.047] on ResNet-50. Spread also improved on the default peak confidence of the same model, by +0.019 [0.016, 0.023] and +0.027 [0.022, 0.032], so the default score is not the best one the head already provides. On the heatmap model, the default peak confidence was about level with spread, with both above soft MC-Dropout by +0.079

Table 2: Zero-shot uncertainty on the test split, with bootstrap 95% intervals on failure AUROC. Bold marks the best value per column within a model. All values use flip-test predictions, so "1 pass" counts what the signal itself requires: *these need the second, flipped pass.

Signal	Passes	Failure AUROC	AUSE ($\times 10^3$)	Occl. AUROC	Spearman ρ	ECE
SimCC-ViPNAS-MBV3						
Peak confidence	1	0.893 [0.886, 0.900]	8.25	0.780	0.549	0.008
Peak probability	1	0.904 [0.897, 0.912]	6.65	0.797	**0.677**	**0.005**
Distribution spread	1	**0.913** [0.907, 0.919]	**5.90**	0.798	0.660	0.023
Entropy	1	0.909 [0.902, 0.916]	6.33	0.802	0.675	0.013
Flip disagreement	2*	0.817 [0.807, 0.827]	13.10	0.719	0.445	0.016
MC-Dropout, argmax (T=20)	20	0.810 [0.799, 0.820]	13.08	0.744	0.524	0.022
MC-Dropout, soft (T=20)	20	0.820 [0.810, 0.830]	11.22	0.741	0.488	0.020
Fusion (logistic)	2*	0.911 [0.904, 0.918]	6.29	**0.803**	0.664	0.012
SimCC-ResNet-50						
Peak confidence	1	0.887 [0.879, 0.895]	9.16	0.757	0.473	0.011
Peak probability	1	0.899 [0.890, 0.909]	6.44	0.795	**0.683**	**0.005**
Distribution spread	1	**0.914** [0.907, 0.921]	**5.60**	0.801	0.661	0.024
Entropy	1	0.908 [0.899, 0.916]	6.03	0.802	0.680	0.012
Flip disagreement	2*	0.819 [0.808, 0.831]	12.08	0.734	0.460	0.013
MC-Dropout, argmax (T=20)	20	0.828 [0.816, 0.840]	11.53	0.740	0.500	0.036
MC-Dropout, soft (T=20)	20	0.873 [0.863, 0.882]	8.88	0.755	0.521	0.030
Fusion (logistic)	2*	0.912 [0.904, 0.920]	6.03	**0.802**	0.659	0.015
Heatmap-ResNet-50						
Peak confidence	1	0.906 [0.898, 0.914]	6.77	0.807	**0.643**	0.013
Distribution spread	1	0.905 [0.896, 0.913]	7.10	**0.809**	0.628	0.015
Entropy	1	0.900 [0.890, 0.908]	7.46	0.803	0.628	0.023
Flip disagreement	2*	0.762 [0.750, 0.773]	15.83	0.679	0.335	0.050
MC-Dropout, argmax (T=20)	20	0.791 [0.778, 0.802]	14.81	0.705	0.362	0.037
MC-Dropout, soft (T=20)	20	0.827 [0.815, 0.839]	13.76	0.744	0.344	0.029
Fusion (logistic)	2*	**0.907** [0.899, 0.915]	**6.70**	0.807	0.641	**0.004**

[0.073, 0.087] and +0.078 [0.071, 0.085]. So the spread's advantage over peak confidence is specific to the SimCC results here, whereas the gap to MC-Dropout holds across all three models. Figure 1 shows the same order across retained fractions. Logistic fusion did not improve on the best single signal in failure AUROC on SimCC, and its intervals overlap those of the best single signal on the heatmap model.

Dropout rate. The planned rate of 0.1 could have been a poor choice, so we repeated inference at $p \in \{0.05, 0.2, 0.3, 0.5\}$ (exploratory). The best MC-Dropout result over all rates and both decoders was 0.844 on ViPNAS-MobileNetV3 (argmax, p=0.5), 0.874 on ResNet-50 (soft, p=0.2), and 0.827 on the heatmap model (soft, p=0.1). All are below the corresponding best one-pass signal (Table 3, Figure 2). No rate closes the gap.

Flip testing. Table 2 uses flip-test predictions, so its one-pass signals in fact come from two backbone passes. With flip testing off (exploratory), spread reached 0.912 and 0.912 on SimCC, and peak confidence 0.901 on the

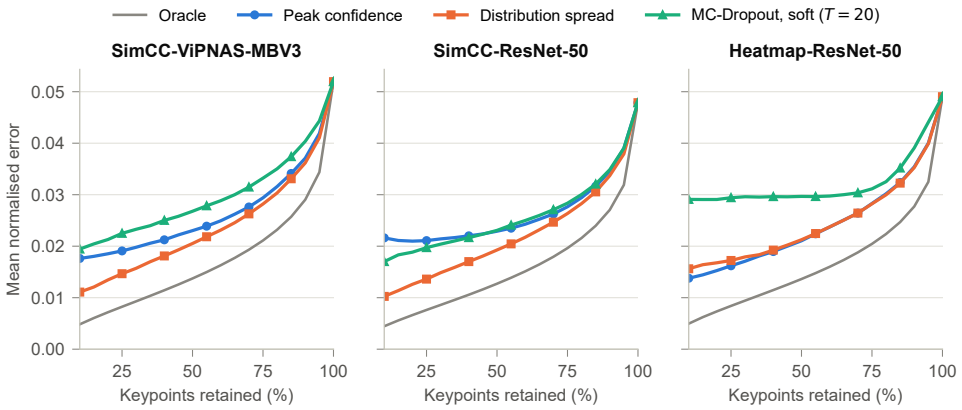


Figure 1: Sparsification on the test split: mean error of the keypoints retained after discarding the most uncertain ones. Lower is better, and the oracle sorts by the true error. Values are in Table 2 (AUSE).

Table 3: Failure AUROC of soft MC-Dropout (T=20) by dropout rate, against one-pass spread. Exploratory. Figure 2 also shows the argmax decoder.

Model	p=0.05	p=0.1	p=0.2	p=0.3	p=0.5	Spread (1 pass)
SimCC-ViPNAS-MBV3	0.817	0.820	0.822	0.823	0.822	0.913
SimCC-ResNet-50	0.873	0.873	0.874	0.873	0.871	0.914
Heatmap-ResNet-50	0.819	0.827	0.812	0.782	0.699	0.905

heatmap model, against 0.824, 0.877 and 0.825 for soft MC-Dropout, so the one-pass claim holds in the strict sense too.

Occlusion. Occlusion AUROC follows the same pattern, at a lower level. The best signal reached 0.803, 0.802 and 0.809 on the three models, compared with 0.741, 0.755 and 0.744 for soft MC-Dropout. Figure 3 shows instances selected by a fixed rule, not by hand.

Conformal regions. Table 4 gives split-conformal results. All variants reach close to the 90% marginal target. A constant radius covered only 0.657 and 0.658 of occluded joints on SimCC. Scaling by spread raised this to 0.770 and 0.762, and changed the mean radius by −13% and −14%. On the heatmap model it raised occluded coverage to 0.875, but at the price of a +47% larger radius, so the benefit there is a trade-off and not a clear gain. Occluded coverage remains below the marginal target in every case: this marginal guarantee does not extend to the occluded subgroup.

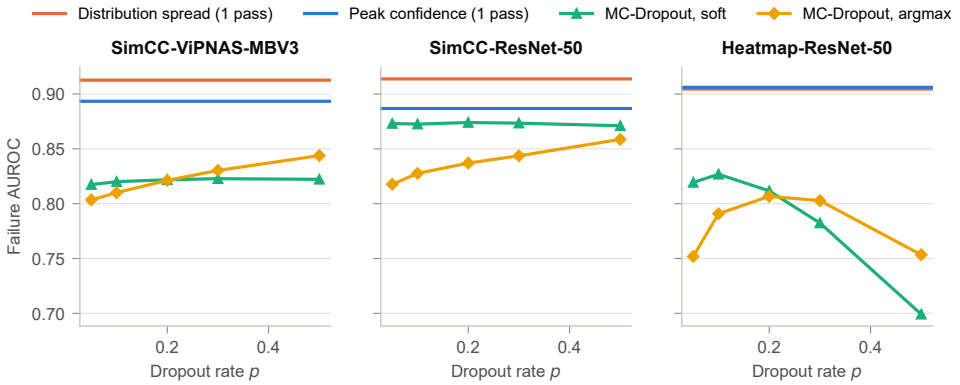


Figure 2: Failure AUROC against dropout rate. One-pass signals are flat lines because they do not depend on the rate.

Table 4: Split-conformal regions at 90% target coverage. Radii are in normalised error units.

Model	Radius scaled by	Coverage	Vis. cov.	Occl. cov.	Mean radius
SimCC-ViPNAS-MBV3	Constant	0.896	0.929	0.657	0.098
	Distribution spread	0.901	0.919	0.770	0.085
	MC-Dropout, soft (T=20)	0.900	0.924	0.719	0.091
SimCC-ResNet-50	Constant	0.898	0.931	0.658	0.091
	Distribution spread	0.899	0.918	0.762	0.078
	MC-Dropout, soft (T=20)	0.897	0.918	0.741	0.085
Heatmap-ResNet-50	Constant	0.896	0.932	0.641	0.090
	Distribution spread	0.902	0.906	0.875	0.132
	MC-Dropout, soft (T=20)	0.900	0.910	0.822	0.135

4.3. RQ2: learned heads

The control, and MC-Dropout after training. The `base` arm fine-tunes the ViPNAS-MobileNetV3 model for 20 epochs with KL only and head dropout active. Its AP on all val2017 images was 0.714 with ground-truth boxes (pretrained: 0.711) and 0.694 with detector boxes (pretrained: 0.691), so the fine-tuning schedule itself does no harm. Training with dropout did help MC-Dropout: failure AUROC rose from 0.820 to 0.859 for the soft decoder and from 0.810 to 0.835 for argmax, and occlusion AUROC from 0.741 to 0.762. The pretrained comparison in Table 2 is therefore somewhat unfavourable to MC-Dropout. The ranking did not change: spread reached 0.912 on the same model, and soft MC-Dropout minus spread was −0.053 [−0.058, −0.047]. This is one seed on one backbone.

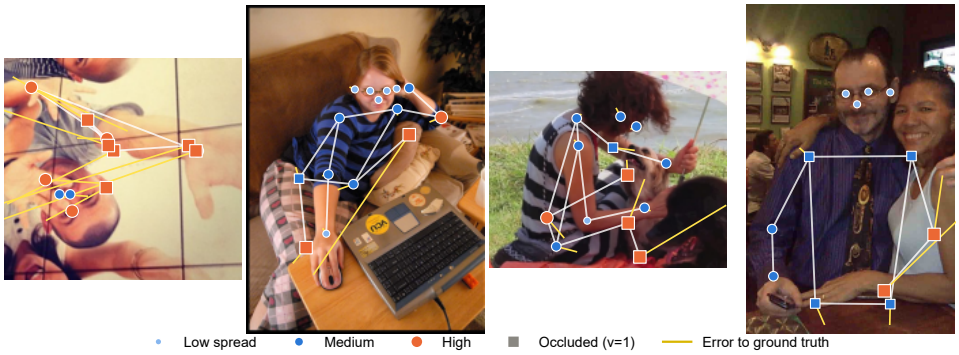


Figure 3: SimCC ViPNAS-MobileNetV3 on test instances chosen by rule: of those with at least 3 occluded and 6 visible labelled joints and a box side of at least 150 pixels, the four with the largest gap between occluded and visible mean error. These are selected failures and do not show typical accuracy.

The full learned head. The `full` arm adds the heteroscedastic, calibration and OKS terms, the token mixer and the adaptive target. Table 5 compares it with the control. On ViPNAS-MobileNetV3 with seed 0 (other seeds below), the learned signals did not beat spread computed on the same model's output. Learned σ minus spread was -0.010 [-0.013, -0.007] in failure AUROC, $+0.58$ [0.41, 0.78] in AUSE ($\times 10^{-3}$, lower is better) and $+0.001$ [-0.001, 0.003] in occlusion AUROC, so σ was worse on the first two and not distinguishable on the third. The OKS head was further behind, at -0.046 [-0.051, -0.041] in failure AUROC. Adding both to logistic fusion did not separate it from spread alone either ($+0.000$ [-0.001, 0.002]).

The head also cost accuracy. On val-test, AP with ground-truth boxes was 0.708 against 0.720 for the control, a mean paired fold difference of -1.24 ± 0.56 AP points, positive in 0 of 10 folds. With detector boxes the difference was -1.03 ± 0.40 points. So the deficit seen at the interim check did not close by the final epoch. The median logged gradient norm before clipping, where each logged value is a 50-iteration mean, was 111.3 for `full` and 0.3 for the control, against a clip threshold of 1.0.

Stopping the pose gradient (exploratory). The `full_sg` arm detaches D in $\mathcal{L}_{\text{het}}$ and is otherwise identical to `full`. Its logged gradient norm stayed large, with a median of 85.3, so the detach did not remove the large norms that prompted this arm. On val-test its AP was 0.715 with ground-truth boxes, -0.38 ± 0.34 points from the control and $+0.86 \pm 0.48$ from `full` (positive in 10 of 10 folds). With detector boxes these were -0.40 ± 0.27 and $+0.63 \pm 0.42$ (positive in 10 of 10 folds). So on this seed the detach removed

Table 5: Fine-tuning arms, final-epoch checkpoints. AP is on val-test images. Δ is the mean paired difference from the control across 10 disjoint val-test folds, in AP points, with the SD across folds, for ground-truth (GT) and detector (det.) boxes. [§]Exploratory arm, added after an interim look at evaluation data (Section 3.5).

		AP, val-test			Failure AUROC			Occlusion AUROC	
Backbone	Arm	GT	Δ GT	Δ det.	Spread	Learned σ	Learned OKS	Spread	Learned σ
ViPNAS-MBV3	Pretrained	0.717	-0.24 ± 0.22	-0.18 ± 0.14	0.913	–	–	0.798	–
	Control (KL only)	0.720	–	–	0.912	–	–	0.799	–
	+ σ head	0.709	-1.09 ± 0.60	-0.84 ± 0.59	0.912	0.907	–	0.799	0.800
	+ OKS head	0.710	-0.94 ± 0.47	-0.86 ± 0.43	0.912	0.907	0.868	0.798	0.799
	+ Token mixer	0.713	-0.75 ± 0.48	-0.54 ± 0.29	0.909	0.903	0.868	0.799	0.800
	+ Adaptive target	0.708	-1.24 ± 0.56	-1.03 ± 0.40	0.913	0.904	0.868	0.796	0.798
	Same, stop-gradient[§]	0.715	-0.38 ± 0.34	-0.40 ± 0.27	0.909	0.905	0.868	0.796	0.799
ResNet-50	Pretrained	0.741	$+0.04 \pm 0.19$	-0.07 ± 0.21	0.914	–	–	0.801	–
	Control (KL only)	0.741	–	–	0.915	–	–	0.801	–
	+ Adaptive target	0.740	-0.26 ± 0.31	-0.15 ± 0.39	0.914	0.898	0.857	0.801	0.796

most of the AP loss, though not all of it. It did not change the ranking of the signals. Learned σ minus spread was -0.005 [-0.007, -0.002] in failure AUROC and $+0.33$ [0.19, 0.51] in AUSE, and only on occlusion AUROC did σ come out ahead, by $+0.003$ [0.001, 0.004]. That occlusion difference is small. An evaluation-set number prompted this arm and it has one seed, so the AP recovery is a hypothesis for later work and not an established result.

ResNet-50. The larger backbone repeats the signal result but not the size of the AP cost, on one seed. Learned σ minus spread was -0.017 [-0.021, -0.013] in failure AUROC, $+0.71$ [0.55, 0.92] in AUSE and -0.005 [-0.007, -0.002] in occlusion AUROC, so here σ was worse on all three. The OKS head was at -0.057 [-0.064, -0.051] in failure AUROC, and fusion with both learned signals was not separated from spread (-0.001 [-0.003, 0.000]). The AP difference from the control was -0.26 ± 0.31 points with ground-truth boxes and -0.15 ± 0.39 with detector boxes, each smaller than its SD across folds, so we cannot call it a cost on this backbone. Training with dropout again helped MC-Dropout, from 0.873 to 0.882 for the soft decoder, but it stayed behind spread (-0.033 [-0.038, -0.028]).

Which component costs AP. The planned arms on ViPNAS-MobileNetV3 add one component at a time, on one seed. The first step carried most of the loss: `het`, which adds only the σ head with $\mathcal{L}_{\text{het}}$ and $\mathcal{L}_{\text{cal}}$, differed from the control by -1.09 ± 0.60 points with ground-truth boxes (positive in 0 of 10 folds) and -0.84 ± 0.59 with detector boxes. Measured directly against the arm before each, with ground-truth boxes, the OKS head changed AP by $+0.14 \pm 0.37$ (positive in 5 of 10 folds), the token mixer by $+0.20 \pm 0.32$ (8 of 10) and the adaptive target by -0.49 ± 0.36 (1 of 10). With detector boxes the adaptive-target step was -0.49 ± 0.46 (1 of 10). So the heteroscedastic

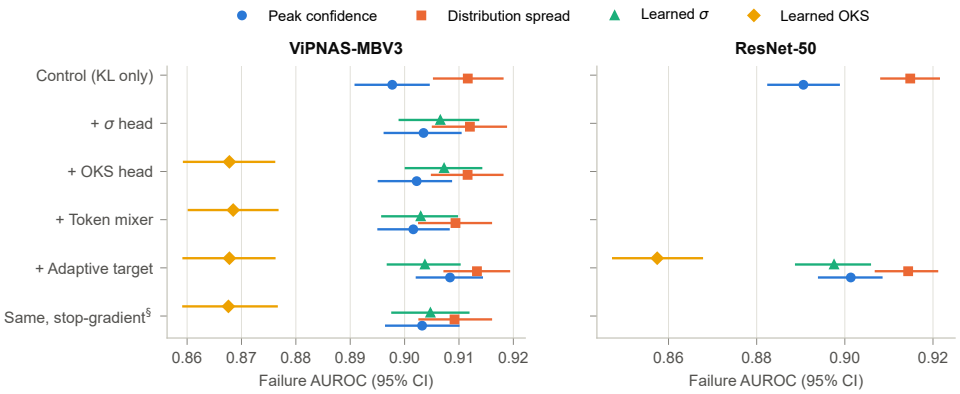


Figure 4: Failure AUROC on val-test for each fine-tuning arm, one seed, with 95% bootstrap intervals. Intervals from different signals on the same images are not independent, so paired differences in the text are the proper comparison. §Exploratory arm.

and calibration terms are the main source of the cost, the OKS head shows no effect, and the adaptive target may add to it. These later steps are about the size of their fold SD, and folds capture image sampling but not variation between training runs, so they are weak evidence on one seed.

No arm brought learned σ level with spread on failure AUROC. The difference was -0.006 [-0.008, -0.003] for `het`, -0.004 [-0.007, -0.002] for `het_oks` and -0.006 [-0.009, -0.004] for `mix`. On occlusion AUROC these three arms could not be separated from spread ($+0.001$ [-0.001, 0.003], $+0.001$ [-0.001, 0.003] and $+0.000$ [-0.002, 0.002]). The added components did not improve the signal either: σ scored 0.907 in the simplest arm and 0.904 in `full`.

Repeated training runs. We trained `base` and `full` on ViPNAS-MobileNetV3 again with seeds 1 and 2 (Table 6). Each seed's AP difference is the paired fold mean against the control trained with the same seed, so the spread across seeds reflects variation between training runs, which the fold SD does not capture. The AP cost appeared on every seed: -1.07 ± 0.16 points with ground-truth boxes (mean and SD over 3 seeds, range -1.24 to -0.91) and -0.86 ± 0.28 with detector boxes (range -1.03 to -0.54). The control itself was stable, at 0.720, 0.720 and 0.719. The signal ranking also held. Learned σ minus spread in failure AUROC was -0.008 ± 0.001 across seeds, with the 95% interval below zero in 3 of 3 seeds. For the OKS head it was -0.043 ± 0.003, below zero in 3 of 3. On occlusion AUROC, σ minus spread averaged $+0.001$, with the interval above zero in 0 of 3 seeds and below zero in 0 of 3.

Table 6: Seeds 0, 1 and 2 of the control and the full head on ViPNAS-MobileNetV3, mean ± SD across seeds. AP is on val-test with ground-truth boxes. Δ is each seed's mean paired fold difference from the control of the same seed, in AP points, then averaged across seeds. σ and OKS are the learned head's outputs. AUROC values are on val-test.

		AP, val-test, GT		Failure AUROC			Occl. AUROC
Arm	Seeds	AP	Δ	Spread	σ	OKS	Spread
Control (KL only)	3	0.7198 ± 0.0004	–	0.911 ± 0.000	–	–	0.798 ± 0.000
Full head	3	0.7096 ± 0.0017	−1.07 ± 0.16	0.911 ± 0.002	0.903 ± 0.001	0.868 ± 0.001	0.797 ± 0.000

Table 7: Model cost at batch size 1, median over 300 runs. "Ratio" is T=20 time over single-pass time on the GPU. ‡Includes the unused uncertainty modules of our head. Sub-10 ms GPU differences should be read with caution, since desktop applications shared the GPU.

	Params	FLOPs	GPU (ms)			CPU, 1 thread (ms)	
Model	(M)	(G)	1 pass	T=20	Ratio	1 pass	T=20
ViPNAS-MBV3, plain SimCC	8.65	0.80	10.7	–	–	49	–
ViPNAS-MBV3, + σ/OKS head	9.06	0.80	10.2	27.8	2.7×	50	107
ViPNAS-MBV3, + token mixer	9.62	0.82	12.6	41.8	3.3×	50	128
ResNet-50, plain SimCC	36.75	5.50	4.9	–	–	123	–
ResNet-50, + σ/OKS head	37.16	5.50	5.5	21.9	4.0×	122	854
ResNet-50, + token mixer	37.72	5.51	8.1	39.8	4.9×	123	906
ResNet-34, SimCC‡	28.65	4.77	5.0	20.6	4.1×	98	607
SOCA-PRNet-34, SimCC‡	28.95	4.77	6.9	23.9	3.5×	101	606
ResNet-50, heatmap‡	34.30	5.45	5.4	28.0	5.2×	138	1235

Three seeds give a rough estimate of run-to-run variation, not a precise one, and the ablation arms, `full_sg` and ResNet-50 remain single runs.

4.4. RQ3: cost

Table 7 gives measured costs. The learned head adds little in FLOPs: from 0.80 G to 0.82 G on ViPNAS-MobileNetV3 including the mixer. Its parameter overhead matters more on the lightweight model, from 8.65 M to 9.62 M. Head-only MC-Dropout with T=20 costs far less than 20 forward passes because the backbone runs once: 2.7× to 5.2× a single pass on the GPU, and 2.1× to 8.9× on one CPU thread. So the case against head MC-Dropout here rests mainly on its weaker signal and not on latency, and any argument based on an assumed 20× cost overstates the difference. The AP cost of the learned head is reported with RQ2.

4.5. RQ4: instance rescoring

Table 8 shows small AP changes. Exponential decay on spread gained +0.16 and +0.26 AP points with ground-truth boxes, and +0.14 and +0.22 with detector boxes, with fold standard deviations of similar size. Expected OKS changed AP by +0.14 and +0.22 points with ground-truth boxes, and

by -0.08 and -0.01 with detector boxes. On the heatmap model, calibration selected $\alpha = 0$ for both box types, meaning no rescoring. Spread-based rescoring is therefore not a meaningful source of AP on these models. Its value lies in identifying which keypoints to distrust.

The fine-tuned `full` heads add two rules that use the learned outputs, expected OKS from σ and the predicted OKS itself, with parameters again selected on val-cal. On ViPNAS-MobileNetV3 neither helped: expected OKS from σ changed AP by -0.06 and -0.09 points with ground-truth and detector boxes, and the predicted OKS by -0.28 ± 0.16 and -0.30 ± 0.26 (positive in 0 of 10 folds). On ResNet-50, expected OKS from σ gained $+0.27 \pm 0.12$ points with ground-truth boxes and $+0.45 \pm 0.11$ with detector boxes (positive in 10 of 10 folds), the only rescoring gain in this study that clearly exceeds its fold SD. It comes from one training run, among several rules and models, so we treat it as a lead to test with more seeds and not as an established gain. It is also smaller than the AP that the head cost on the lightweight backbone.

4.6. SOCA-PRNet

Table 9 compares the two backbones trained with the 60-epoch schedule of Section 3.5. SOCA attention adds 28.95 M against 28.65 M parameters, with FLOPs unchanged at two decimals (4.77 G and 4.77 G). On val-test, AP with ground-truth boxes was 0.715 against 0.712, a mean paired fold difference of $+0.44 \pm 0.53$ points (positive in 9 of 10 folds). With detector boxes it was $+0.48 \pm 0.56$ (positive in 9 of 10 folds). The difference leans towards SOCA-PRNet, but each mean is smaller than its SD across folds, and with one training run per backbone the variation between runs is not measured. Our fine-tuning seeds give only a rough guide to that variation, since training from ImageNet weights for 60 epochs is a different setting. We therefore read this as a small possible gain, not an established one. The uncertainty result is the same on both backbones: spread reached a failure AUROC of 0.913 on SOCA-PRNet-34 and 0.910 on ResNet-34, against 0.897 and 0.892 for peak confidence.

5. Discussion

Why spread beats head MC-Dropout here. We offer explanations, but did not test them causally. The spread summarises ambiguity that the trained classifier has already learned to express: an occluded joint tends to produce a wide or multi-modal distribution. Head-only dropout, by contrast, perturbs a few layers around one fixed backbone feature, and a pretrained model was

Table 8: Instance rescoring on the test split, with parameters selected on the calibration split, for the pretrained SimCC models and the fine-tuned full heads (seed 0). All models use SimCC. ΔAP is in percentage points, paired over 10 folds. Test AP differs from Table 1 because it covers the test images only.

Model	Boxes	Rule	Test AP	ΔAP (mean ± SD, 10 folds)	Folds > 0
ViPNAS-MBV3, pretrained	GT	Default	0.717	–	–
		Exp. decay (spread)	0.719	+0.16 ± 0.17	9/10
		E[OKS] (spread)	0.719	+0.14 ± 0.09	9/10
	Det.	Default	0.698	–	–
		Exp. decay (spread)	0.700	+0.14 ± 0.20	9/10
		E[OKS] (spread)	0.697	−0.08 ± 0.37	5/10
ResNet-50, pretrained	GT	Default	0.741	–	–
		Exp. decay (spread)	0.744	+0.26 ± 0.26	9/10
		E[OKS] (spread)	0.743	+0.22 ± 0.15	10/10
	Det.	Default	0.724	–	–
		Exp. decay (spread)	0.726	+0.22 ± 0.30	8/10
		E[OKS] (spread)	0.724	−0.01 ± 0.30	4/10
ViPNAS-MBV3, full head	GT	Default	0.708	–	–
		Exp. decay (spread)	0.709	+0.08 ± 0.16	9/10
		E[OKS] (spread)	0.707	−0.08 ± 0.09	3/10
		E[OKS] (learned σ)	0.707	−0.06 ± 0.12	4/10
		Learned OKS	0.705	−0.28 ± 0.16	0/10
	Det.	Default	0.690	–	–
		Exp. decay (spread)	0.691	+0.09 ± 0.20	8/10
		E[OKS] (spread)	0.686	−0.37 ± 0.37	1/10
		E[OKS] (learned σ)	0.689	−0.09 ± 0.32	5/10
		Learned OKS	0.687	−0.30 ± 0.26	0/10
ResNet-50, full head	GT	Default	0.740	–	–
		Exp. decay (spread)	0.742	+0.26 ± 0.12	10/10
		E[OKS] (spread)	0.741	+0.12 ± 0.09	10/10
		E[OKS] (learned σ)	0.743	+0.27 ± 0.12	10/10
		Learned OKS	0.741	+0.14 ± 0.10	10/10
	Det.	Default	0.723	–	–
		Exp. decay (spread)	0.725	+0.14 ± 0.13	9/10
		E[OKS] (spread)	0.721	−0.24 ± 0.30	1/10
		E[OKS] (learned σ)	0.728	+0.45 ± 0.11	10/10
		Learned OKS	0.726	+0.23 ± 0.14	9/10

not fit under this perturbation. Training with the dropout active narrowed the gap but did not close it (Section 4.3), which supports the training part of this account only partly. The remainder may reflect that head dropout around one fixed backbone feature does not reach the sources of error that matter, a possibility we did not test.

Why the learned head did not help. Again we did not test the explanation. A SimCC head is already trained to put probability where the keypoint is, so its spread already carries much of what a separate σ head is asked to learn, and σ is fit to the same training errors. The learned outputs would then mostly restate the spread, which matches how close σ came to it on every arm. The AP cost points to an interaction with the pose objective: the arm that added only $\mathcal{L}_{\text{het}}$ and $\mathcal{L}_{\text{cal}}$ carried most of the cost, and detaching the

Table 9: ResNet-34 and SOCA-PRNet-34 with the same SimCC head, trained from ImageNet weights for 60 epochs, one run each. AP is on val-test images. The shortened schedule means these values cannot be compared with published 210-epoch results.

			AP, val-test		Failure AUROC	
Backbone	Params (M)	FLOPs (G)	GT	det.	Spread	Peak conf.
ResNet-34	28.65	4.77	0.712	0.693	0.910	0.892
SOCA-PRNet-34	28.95	4.77	0.715	0.698	0.913	0.897

pose term inside $\mathcal{L}_{\text{het}}$ recovered most of it, though in a single exploratory run. The logged gradient norms stayed large with the detach, so the mechanism is not settled by our data.

Practical reading. On a SimCC model, computing the spread needs no training, no extra parameters and only a small post-processing step on outputs the model already produces. On these models it is a sensible default before adding machinery. On heatmap models, the default score already performs comparably. These findings are about the models and metrics tested, and do not rule out that other methods, such as deep ensembles, would do better.

Limitations. The study uses one dataset, top-down models and ground-truth boxes for keypoint-level metrics, so detector errors are not reflected in them. MC-Dropout is limited to the head. We did not evaluate full-network dropout or ensembles, and the backbones were not trained with dropout. Only the control and the full head on ViPNAS-MobileNetV3 were trained with three seeds. Every other arm, ResNet-50 and the SOCA comparison are single runs, and all training schedules are short. COCO occlusion flags are annotator judgements. The test split is drawn from val2017, which is a public benchmark and may have influenced the published models. Figures quoted at the third decimal come from one split, though bootstrap intervals capture image sampling within it. Our detector differs from the one behind the published numbers.

6. Conclusion

Across three published pose models, signals already present in one pass detected keypoint failures better than 20-pass head MC-Dropout, at every dropout rate we tested, and with or without flip testing. Spread-scaled conformal regions helped on SimCC, and rescoring instances with the spread

did not move AP meaningfully. Fine-tuning a learned uncertainty head did not change this picture. Its σ and OKS outputs ranked failures worse than the spread of the same fine-tuned model, on two backbones and, for the lightweight one, on three seeds, and the head cost about one AP point there. The component that carried most of that cost was the heteroscedastic term itself, and stopping its pose gradient, an exploratory single-seed arm, recovered most of the loss without making σ better than spread. For SimCC models of this kind we would compute the spread first, and add a trained uncertainty head only where a measured gain justifies its parameters and its AP cost.

Appendix A. Defects found and fixed

We report these because each would have changed the results while the code still ran. Regression tests for them are in `tests/test_core.py`.

- **Pretrained heatmap weights failed to load**: inserting dropout as its own layer shifted the deconvolution indices, so those weights were silently left random.
- **Resuming started from random weights**: `-resume auto` cleared the pretrained-checkpoint path even when no checkpoint existed to resume from.
- **The anatomical prior in the mixer was overwritten** by the framework's weight initialisation.
- **ECE was computed on raw logits**, not on probabilities, and on the same data used for fitting.
- **Ties in AUROC and risk-coverage** were ordered arbitrarily, which matters because argmax MC-Dropout often ties at zero.
- **Inconsistent units**: the residual used for calibration and the OKS target were not on the scale their formulas assumed. Unlabelled joints also received an OKS target.

Data and code availability

PENDING, no result yet: Repository URL. COCO is public. Result files regenerate every table, figure and inline number through `scripts/make_paper_assets.py`.

Declaration of competing interest

PENDING, no result yet: To be completed by the authors.

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